Braided-Symmetric Derivations of Algebras

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ABSTRACT
We study braided-symmetric derivations of an algebra over a field. These are maps that satisfy the Leibniz rule for the product symmetrized by an involution of $A \otimes A$. The choice of involution decides which derivations appear. The identity involution gives the ordinary derivations when 2 is invertible. The swap gives the Jordan derivations, and its antisymmetric version gives the Lie derivations. So we ask when a Jordan or Lie derivation is a derivation. This question becomes a comparison of two involutions on one algebra. On full matrix algebras and on upper triangular 2×2 matrices, a trace-type map gives an example. It is a Lie derivation but not a derivation. We then twist the swap by an involutive automorphism. Let the algebra be commutative and reduced, with 2 invertible. Then the twisted maps that commute with the automorphism are exactly the derivations that commute with it. An example with nilpotent elements shows that the reduced hypothesis is needed.

KEYWORDS
Braided-symmetric derivations; twisted Jordan derivations; triangular algebras; Jordan derivations; Lie derivations

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1. Introduction

Throughout, K is a field and an algebra means a unital associative K-algebra. We take all maps between algebras to be K-linear. We say so when a result cited below holds more generally for additive maps on rings.

A *derivation* of an algebra A is a linear map $D\colon A \to A$ with $D(xy) = D(x)y + xD(y)$ for all $x, y \in A$. The notion goes back to Jacobson [14, Part I], who studied the derivations of an arbitrary nonassociative algebra. Hochschild [12, §§2 and 4] defined the cocycles of an associative algebra, and its 1-cocycles are the derivations. A *Jordan derivation* of A is a linear map with $D(x \circ y) = D(x) \circ y + x \circ D(y)$ for all $x, y \in A$, where $x \circ y = xy + yx$. Herstein [10, p. 1104] introduced this notion. His result needs three notions for a ring R. We call R *prime* if $aRb = 0$ implies $a = 0$ or $b = 0$ [5, p. 1003]. We call R *semiprime* if $aRa = 0$ implies $a = 0$ [5, p. 1003]. We call R 2-*torsion free* if $2a = 0$ implies $a = 0$ [5, p. 1003]. So every prime ring is semiprime. Herstein proved that every Jordan derivation of a prime ring of characteristic different

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from 2 is a derivation [10, Theorem 3.1]. Every derivation is a Jordan derivation, and the converse is false in general [5, p. 1003]. Cusack [9, Corollary 5] extended this result of Herstein to 2-torsion free semiprime rings, and Brešar [5, Theorem 1] gave an independent proof.

A *Lie derivation* of A is a linear map with $D([x,y]) = [D(x),y] + [x,D(y)]$, where $[x,y] = xy - yx$. Martindale [18, p. 183] studied the Lie derivations of a primitive ring of characteristic not 2 that contains a nontrivial idempotent, and showed that each has the form $D+T$, with D a derivation and T a map into the centre that vanishes on commutators. So for Lie derivations no result of Herstein type can hold. Here is the reason. Let $t\colon A \to Z(A)$ be a linear map into the centre with $t([x,y]) = 0$ for all x, y. Every term of the Lie derivation identity for t is then zero, so t is a Lie derivation. But t is not a derivation as soon as $t(1) \neq 0$.

So a description of Lie derivations must allow such a term. A Lie derivation has the *standard form* if it equals $d + t$, with d a derivation and t a central-valued map that vanishes on commutators [8, §1]. A ring is *left primitive* if it has a faithful simple left module, and right primitive rings are defined in the same way [17, Definition 11.2]. Martindale [18, Theorem 2] proved that every Lie derivation has this form for primitive rings of characteristic not 2 with a nontrivial idempotent. Here an *idempotent* is an element e with $e^2 = e$, and it is *nontrivial* if $e \neq 0$ and $e \neq 1$. Here d may take values in a larger ring.

The next result uses the *standard polynomial* of degree 4 [2, §1],

$$S_4(x_1,x_2,x_3,x_4) = \sum_{\varpi} \operatorname{sgn}(\varpi)\, x_{\varpi(1)}x_{\varpi(2)}x_{\varpi(3)}x_{\varpi(4)},$$

where ϖ runs over the permutations of $\{1,2,3,4\}$. We say that a ring *satisfies* S_4 if S_4 is zero for every choice of four elements of the ring. A prime ring satisfies S_4 exactly when it is commutative or is isomorphic to a subring of $M_2(F)$ for a field F [6, Lemma 1]. Brešar [6, Theorem 5] proved the standard form for prime rings of characteristic different from 2 that do not satisfy S_4.

Triangular algebras are defined in Section 2. For them, Zhang and Yu [22, Theorem 2.1] proved that every Jordan derivation of a triangular algebra is a derivation. For Lie derivations, Cheung [8, Theorem 11] found conditions under which every Lie derivation has the standard form. Li and Wei [16, §3] extended this to more algebras. Wang [23, Theorem 3.1 and Corollary 3.1] treated every triangular algebra, with no condition on the module or the centre. For these algebras a Lie derivation is a sum $\delta + \tau$, where δ is a derivation into a larger triangular algebra and τ is a linear map into its centre that vanishes on commutators.

In the result of Wang, every Lie derivation of a triangular algebra $\mathcal{U}$ is a sum $d + t$ [23, Theorem 3.1]. Here d is a derivation of $\mathcal{U}$ into a larger triangular algebra $\mathcal{U}_0$ with the same unit [23, Theorem 3.1]. The map t is linear, takes values in the centre of $\mathcal{U}_0$ and vanishes on commutators [23, Theorem 3.1].

Benkovič [3, §1] treats derivations, Jordan derivations and Lie derivations as f-derivations, for $f = x_1x_2$, $x_1 \circ x_2$ and $[x_1,x_2]$, so that one identity for a multilinear polynomial f covers all three. We obtain them from an identity of a different kind. Our identity keeps the polynomial x_1x_2, and an involution of $A \otimes A$ acts before we multiply. Here the involution may use extra structure of A, such as an automorphism. The identity is then no longer an f-derivation identity; Section 2 explains why.

Let A be an algebra over K with product μ. Let $\sigma\colon A \otimes A \to A \otimes A$ be a K-linear map with $\sigma^2 = \mathrm{id}$, for instance the swap $\tau(x \otimes y) = y \otimes x$. It gives a symmetrized

product

$$x \cdot_\sigma y = \mu(x \otimes y) + \mu(\sigma(x \otimes y)).$$

For $\sigma = \tau$ this is the Jordan product $x \circ y$. Put $\mu_\sigma = \mu + \mu \circ \sigma$. Since $\sigma^2 = \mathrm{id}$, this gives

$$\mu_\sigma \circ \sigma = \mu_\sigma.$$

Example 3.3 shows that a twisted version of this product can fail to be associative and to have a unit.

We call a linear map $D\colon A \to A$ a *braided-symmetric derivation* for σ if

$$D(x \cdot_\sigma y) = D(x) \cdot_\sigma y + x \cdot_\sigma D(y). \tag{1}$$

So (1) is the Leibniz rule for the symmetrized product. Other authors have studied *braided derivations*, which use a braiding operator directly [20, §2] [15, §4.1, Definition 14] [21, §3, equation (3.9)]. Remark 3.2 compares the two identities. Our identity uses only μ, σ and D.

We now describe the main results. Our first step is a dictionary. Remark 3.4 shows that the identity gives exactly the derivations, and the swap gives exactly the Jordan derivations.

Our second step twists the swap by an involutive automorphism or anti-automorphism α of A. Proposition 4.2 describes the twisted class through the defect $E_D(x, y) = D(xy) - D(x)y - xD(y)$. Corollary 4.5 shows that, on 2-torsion-free semiprime rings, the Jordan maps in the twisted class are the derivations commuting with α.

Theorem 5.1 treats commutative reduced algebras and needs no Jordan hypothesis. Example 5.2 shows that the reduced hypothesis is needed. Proposition 5.4 shows that the twisted class and the α-derivations on one algebra do not contain each other. Proposition 3.11 shows that these spaces keep their dimension when the base field grows.

Our third step uses the difference $\mu - \mu \circ \sigma$ in place of the sum. Proposition 6.2 shows that it gives the Lie derivations, together with the easy half of the theorem of Martindale and Brešar. Proposition 6.3 describes the twisted version through the same defect. Example 6.9 finds both spaces for full matrix algebras. Corollaries 3.7 and 6.11 apply the results to triangular algebras, where the theorems of Zhang and Yu and of Cheung give complete answers. Example 6.12 checks them on the smallest case.

We follow this order. Section 2 recalls the classical notions and contains no new results. Section 3 defines braided-symmetric derivations and treats the identity and the swap. Section 4 treats the twisted swap through the Leibniz defect. Section 5 treats reduced commutative algebras and compares the twisted class with α-derivations. Section 6 treats the antisymmetrized product. Triangular algebras appear as corollaries in Sections 3 and 6. We place each example right after the result it illustrates.

2. Preliminaries

This section recalls the classical notions we need, and none of it is new. We treat five kinds of derivation and then triangular algebras.

From now on, A is a K-algebra with unit 1. We write $[x, y] = xy - yx$ for the commutator, $x \circ y = xy + yx$ for the Jordan product, and $Z(A)$ for the centre. The centre is the set of elements z with $zx = xz$ for all $x \in A$. We call an element x *nilpotent* if $x^n = 0$ for some $n \geq 1$. A commutative algebra is *reduced* if its only nilpotent element is zero [7, 1.4.12].

A derivation of an algebra is as defined by Jacobson [14, Part I]: a linear map D with $D(xy) = D(x)y + xD(y)$. Every derivation satisfies $D(1) = 0$, since $D(1) = D(1 \cdot 1) = 2D(1)$. For $g \in A$, the map $\mathrm{ad}_g(x) = [g, x]$ is an *inner derivation*. We write $\mathrm{Der}(A)$ and $\mathrm{Inn}(A)$ for these two spaces.

We add a word on the characteristic for Jordan derivations. A *Jordan derivation* in the sense of Herstein [10, p. 1104] is an additive map with $D(x^2) = D(x)x + xD(x)$. We write $\mathrm{JDer}(A)$ for the space of linear maps with $D(x \circ y) = D(x) \circ y + x \circ D(y)$ for all x, y. Replacing x by $x+y$ gives the condition $D(x \circ y) = D(x) \circ y + x \circ D(y)$ in every characteristic. Setting $y = x$ in this condition gives $2D(x^2) = 2(D(x)x + xD(x))$. So the two conditions agree when 2 is invertible in K, and we use the $\circ$ form throughout. When $\mathrm{char}\, K = 2$ and A is commutative, $x \circ y = 0$, and every linear map satisfies the $\circ$ condition.

We need two more kinds. A *Lie derivation* satisfies $D([x, y]) = [D(x), y] + [x, D(y)]$ [18, p. 183], and we write $\mathrm{LDer}(A)$ for this space. When A is commutative, every bracket is zero, so every linear map is a Lie derivation. For an automorphism α of A, an *α-derivation* satisfies $D(xy) = D(x)\alpha(y) + \alpha(x)D(y)$ [13, §1]. We write $\mathrm{Der}_\alpha(A)$ for this space.

We now compare these spaces. When 2 is invertible, $xy = \frac{1}{2}(x \circ y) + \frac{1}{2}[x, y]$. A derivation respects xy, so it respects both halves, and $\mathrm{Der}(A) \subseteq \mathrm{JDer}(A) \cap \mathrm{LDer}(A)$. Both inclusions can be strict, as Examples 3.5 and 6.9 show.

Section 1 recalled the results of Herstein, Cusack and Brešar on Jordan derivations, and the central term t that Lie derivations must allow. In the theorems of Martindale [18, Theorem 2] and Brešar [6, Theorem 5], every Lie derivation has the form $D = d + t$. Here d is a derivation into the central closure. Here the map t is additive, takes values in the extended centroid, and vanishes on commutators.

We now recall these two notions for a prime ring R [6, p. 525]. Take a nonzero ideal U of R and a map $f\colon U \to R$ of right R-modules. We treat two such maps as equal if they agree on a nonzero ideal that lies in both domains. We add and compose these maps, and this gives a ring Q [6, p. 525].

The ring R lies inside Q, since each element a of R acts as the map $x \mapsto ax$ [6, p. 525]. The centre C of Q is a field [6, p. 525]. We call C the *extended centroid* of R. We call the ring $RC + C$ the *central closure* of R. Both notions go back to Martindale [6, p. 525].

We recall a polynomial way to unify these maps. A *multilinear polynomial* in variables $x_1, \ldots, x_n$ that do not commute has the following form.

$$f = \sum_{\varpi} \kappa_\varpi \, x_{\varpi(1)} \cdots x_{\varpi(n)}.$$

Here ϖ runs over the permutations of $\{1, \ldots, n\}$, and the coefficients κ_ϖ lie in K [3, §1, (1.1)].

Fix a nonzero multilinear polynomial f in n variables. An *f-derivation* satisfies the

identity

$$D(f(x_1,\dots,x_n)) = \sum_{i=1}^{n} f(x_1,\dots,D(x_i),\dots,x_n)$$

for all $x_1,\dots,x_n$ [3, §1, (1.2)]. Derivations, Jordan derivations and Lie derivations are the f-derivations for $f = x_1x_2$, $x_1 \circ x_2$ and $[x_1,x_2]$ [3, §1]. In each case f uses only sums, products and scalars, so it makes sense in every algebra.

The twisted products of Section 4 are different. They have the form

$$xy + \alpha(y)\alpha(x)$$

for a fixed automorphism or anti-automorphism α of A. Here the expression applies α to the variables, and α is not a polynomial operation. So it is not of the form $f(x,y)$ for any polynomial f. Hence the results on f-derivations do not apply directly.

Our examples include triangular algebras. An (A,B)-*bimodule* is a left A-module and right B-module M with $(am)b = a(mb)$ for all $a \in A$, $m \in M$ and $b \in B$ [17, Example 1.14]. We call it *unital* if $1m = m = m1$ for all $m \in M$.

Let R be a commutative ring, let A and B be unital R-algebras, and let M be a unital (A,B)-bimodule. The *triangular algebra* $T = \mathrm{Tri}(A,M,B)$ is the set of matrices $\left(\begin{smallmatrix} a & m \\ 0 & b \end{smallmatrix}\right)$ with the matrix sum and product [8, §2]. The module M is *faithful* as a left A-module if $aM = 0$ implies $a = 0$, and as a right B-module if $Mb = 0$ implies $b = 0$. The idempotents $e_1 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $e_2 = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ satisfy $e_2Te_1 = 0$. So T is not prime when $M \neq 0$, and the prime-ring results above do not apply to triangular algebras.

We use three involutions of $A \otimes A$: the identity, the swap $\tau(x \otimes y) = y \otimes x$, and the α-twisted map $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$ of Section 4.

Let $\sigma\colon A \otimes A \to A \otimes A$ be a K-linear map with $\sigma^2 = \mathrm{id}$. We write $x \cdot_\sigma y = xy + \mu(\sigma(x \otimes y))$. We ask for no further axiom. The derivations below are called braided-symmetric because σ plays the role of the braiding operator in the work on braided derivations cited above.

3. Braided-symmetric derivations

This section defines braided-symmetric derivations and identifies them for the identity and the swap.

Definition 3.1. Let A be a K-algebra and let σ be a K-linear involution of $A \otimes A$. A *braided-symmetric derivation* for σ is a K-linear map $D\colon A \to A$ that satisfies identity (1), that is, $D(x \cdot_\sigma y) = D(x) \cdot_\sigma y + x \cdot_\sigma D(y)$ for all $x, y \in A$. We write $\mathrm{BSDer}_\sigma(A)$ for the set of braided-symmetric derivations for σ. This set is a K-subspace of $\mathrm{End}_K(A)$, since the condition is linear in D. For an automorphism α of A, we write $\mathrm{BSDer}_\sigma(A)^\alpha$ for the subset of maps D with $D\alpha = \alpha D$.

The name is close to an older one, and the next remark separates the two.

Remark 3.2. Braided derivations [20, §2] [15, §4.1, Definition 14] form a different notion from braided-symmetric derivations, although both use a braiding. Weber [21, §3, equation (3.9)] defines a braided derivation X of a braided commutative module algebra A over a triangular Hopf algebra (H,R) by a Leibniz rule for the product ab

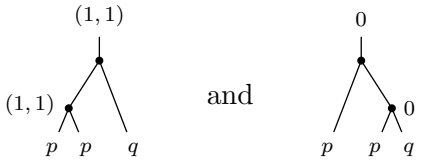
itself:

$$X(ab) = X(a)b + (R_1^{-1} \triangleright a)\big((R_2^{-1} \triangleright X)(b)\big).$$

The braiding enters through the R-matrix. Definition 3.1 asks instead for the ordinary Leibniz rule for the new product $\mu + \mu \circ \sigma$.

Example 3.3 shows that the product $\cdot_\sigma$ can fail to be associative and unital, even when A is commutative.

Example 3.3. Let $A = K \times K$ with componentwise operations, and let $\alpha(a, b) = (b, a)$. Then α is an automorphism with $\alpha^2 = \mathrm{id}$, and $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$ is an involution of $A \otimes A$. Since A is commutative, $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$. For $x = (x_1, x_2)$ and $y = (y_1, y_2)$ this gives $x \cdot_{\sigma_\alpha} y = (x_1y_1 + x_2y_2)(1, 1)$. Put $p = (1, 0)$ and $q = (0, 1)$. Then $(p \cdot_{\sigma_\alpha} p) \cdot_{\sigma_\alpha} q = (1, 1) \cdot_{\sigma_\alpha} q = (1, 1)$, while $p \cdot_{\sigma_\alpha} (p \cdot_{\sigma_\alpha} q) = p \cdot_{\sigma_\alpha} 0 = 0$. The two bracketings are the two trees



where each dot is the product $\cdot_{\sigma_\alpha}$ and each label is the value on that edge. So $\cdot_{\sigma_\alpha}$ is not associative. Every product $u \cdot_{\sigma_\alpha} y$ lies in $K(1, 1)$, so $u \cdot_{\sigma_\alpha} p \neq p$ for every u, and $\cdot_{\sigma_\alpha}$ has no unit.

The next remark records what Definition 3.1 says for the two involutions that use no structure of A.

Remark 3.4. Let A be a K-algebra and let σ be an involution of $A \otimes A$. Identity (1) for σ is the Leibniz rule for the product $\cdot_\sigma$. So $\mathrm{BSDer}_\sigma(A)$ is the set of derivations of the nonassociative algebra $(A, \cdot_\sigma)$, in the sense of Jacobson [14, Part I].

For the identity, $x \cdot_{\mathrm{id}} y = 2xy$, so the identity reads

$$2D(xy) = 2\big(D(x)y + xD(y)\big).$$

If $\operatorname{char} K \neq 2$, we divide by 2 and get $\mathrm{BSDer}_{\mathrm{id}}(A) = \mathrm{Der}(A)$. If $\operatorname{char} K = 2$, then $x \cdot_{\mathrm{id}} y = 0$, so $\mathrm{BSDer}_{\mathrm{id}}(A)$ contains every linear map. So it is larger than $\mathrm{Der}(A)$. For example, take the linear map with $D(1) = 1$ and $D = 0$ on a complement of $K1$. Then D is not a derivation, since a derivation satisfies $D(1) = D(1 \cdot 1) = 2D(1) = 0$ in characteristic 2.

For the swap τ we get $x \cdot_\tau y = xy + yx = x \circ y$. So the defining identity is the $\circ$ form of the Jordan identity from Section 2. Hence $\mathrm{BSDer}_\tau(A) = \mathrm{JDer}(A)$ in every characteristic.

Every derivation is a Jordan derivation [5, p. 1003]. So in every characteristic

$$\mathrm{Der}(A) \subseteq \mathrm{BSDer}_\tau(A) = \mathrm{JDer}(A).$$

When $\operatorname{char} K \neq 2$, this gives $\mathrm{BSDer}_{\mathrm{id}}(A) \subseteq \mathrm{BSDer}_\tau(A)$. This is an inclusion between the spaces of two involutions on one algebra.

The smallest nonreduced algebra shows both inclusions of Section 2 at work.

Example 3.5 (Dual numbers)**.** Let $A = K[\varepsilon]/(\varepsilon^2)$, with basis $\{1, \varepsilon\}$. A linear map D is a derivation exactly when the Leibniz rule holds on the four pairs of basis elements. Since $D(1) = 0$, the pairs $(1, \varepsilon)$ and $(\varepsilon, 1)$ hold. Write $D(\varepsilon) = \beta + c\varepsilon$ with $\beta, c \in K$. The pair $(\varepsilon, \varepsilon)$ gives $0 = D(\varepsilon^2) = 2\varepsilon D(\varepsilon) = 2\beta\varepsilon$. If $\operatorname{char} K \neq 2$, then $\beta = 0$, and every derivation has the form $D(a + b\varepsilon) = bc\varepsilon$, so $\dim_K \operatorname{Der}(A) = 1$. If $\operatorname{char} K = 2$, then β is free, and $\dim_K \operatorname{Der}(A) = 2$. Since A is commutative, every commutator is zero, so $\operatorname{LDer}(A) = \operatorname{End}_K(A)$, which has dimension $2 \cdot 2 = 4$. Each derivation lies in $\operatorname{BSDer}_\tau(A)$ by Remark 3.4. If $\operatorname{char} K = 2$, then $x \circ y = 2xy = 0$ for all x, y, so $\operatorname{JDer}(A) = \operatorname{End}_K(A)$ has dimension 4, which exceeds $\dim_K \operatorname{Der}(A) = 2$.

The remark above and the following theorem on triangular algebras give a corollary.

Theorem 3.6 ([22, Theorem 2.1])**.** *Let R be a 2-torsion free commutative ring, let A and B be unital R-algebras, and let M be a unital (A, B)-bimodule that is faithful as a left A-module and as a right B-module. Then every Jordan derivation $D\colon \operatorname{Tri}(A, M, B) \to \operatorname{Tri}(A, M, B)$ is a derivation.*

In the next corollary, maps on $T = \operatorname{Tri}(A, M, B)$ are R-linear and the braided-symmetric identity is read over R. It follows from Remark 3.4 and Theorem 3.6.

Corollary 3.7. *Let R be a 2-torsion free commutative ring, let A and B be unital R-algebras, and let M be a unital (A, B)-bimodule that is faithful as a left A-module and as a right B-module. Suppose also that $\operatorname{Tri}(A, M, B)$ is 2-torsion free. Then $\operatorname{BSDer}_\tau(\operatorname{Tri}(A, M, B)) = \operatorname{Der}(\operatorname{Tri}(A, M, B))$.*

Proposition 3.11 below treats extension of the base field. For a field extension L of K and a K-vector space V, write $V_L = L \otimes_K V$. This is an L-vector space. For a K-algebra A, the space A_L is an L-algebra with $(c \otimes x)(c' \otimes y) = cc' \otimes xy$ [17, Chapter 2, §5, p. 74].

We now define the involution σ_L of $A_L \otimes_L A_L$. It is an L-linear map of $A_L \otimes_L A_L$ into itself with $\sigma_L^2 = \mathrm{id}$.

Define

$$\Psi\colon A_L \otimes_L A_L \to L \otimes_K (A \otimes_K A), \qquad (c \otimes x) \otimes_L (c' \otimes y) \longmapsto cc' \otimes (x \otimes y).$$

Since this assignment is L-bilinear, it defines an L-linear map. It is an isomorphism, because its inverse sends $c \otimes (x \otimes y)$ to $(c \otimes x) \otimes_L (1 \otimes y)$.

The map $\mathrm{id}_L \otimes \sigma$ is an L-linear map of $L \otimes_K (A \otimes_K A)$ into itself, and $(\mathrm{id}_L \otimes \sigma)^2 = \mathrm{id}_L \otimes \sigma^2 = \mathrm{id}$. We define σ_L as the conjugate of $\mathrm{id}_L \otimes \sigma$ by Ψ:

$$\sigma_L = \Psi^{-1} \circ (\mathrm{id}_L \otimes \sigma) \circ \Psi.$$

For $c, c' \in L$ and $x, y \in A$ it reads

$$\sigma_L\big((c \otimes x) \otimes_L (c' \otimes y)\big) = \Psi^{-1}\big(cc' \otimes \sigma(x \otimes y)\big).$$

Since $\Psi\Psi^{-1} = \mathrm{id}$, we get

$$\sigma_L^2 = \Psi^{-1}(\mathrm{id}_L \otimes \sigma)(\mathrm{id}_L \otimes \sigma)\Psi = \Psi^{-1}(\mathrm{id}_L \otimes \sigma^2)\Psi = \mathrm{id}.$$

So σ_L is an involution of $A_L \otimes_L A_L$.

The product μ_L of A_L satisfies $\mu_L = (\mathrm{id}_L \otimes \mu) \circ \Psi$. So Ψ carries μ_L and σ_L to $\mathrm{id}_L \otimes \mu$ and $\mathrm{id}_L \otimes \sigma$. Hence $(1 \otimes x) \cdot_{\sigma_L} (1 \otimes y) = 1 \otimes (x \cdot_\sigma y)$ for $x, y \in A$. Thus the product $\cdot_{\sigma_L}$ on A_L extends the product $\cdot_\sigma$ on A.

The proof uses three results from the literature, which we recall. For a K-vector space E we write $E_L = L \otimes_K E$, and for a K-linear map f we write $f_L = \mathrm{id}_L \otimes f$.

Theorem 3.8 ([4, Chapter II, §7, no. 9, Proposition 18 and its Corollary])**.** *Let L be an extension field of the field K. If $E' \to E \to E''$ is an exact sequence of K-vector spaces and K-linear maps, then $E'_L \to E_L \to E''_L$ is exact. Hence, for every K-linear map $f\colon E \to F$, we have $\ker(f_L) = (\ker f)_L$ and $\mathrm{im}(f_L) = (\mathrm{im}\, f)_L$, up to canonical isomorphism.*

Theorem 3.9 ([19, Theorem 3.1])**.** *Every free module over a ring is projective.*

Theorem 3.10 ([4, Chapter II, §5, no. 3, Proposition 7(ii)])**.** *Let A be a commutative ring, let B be a ring, let $\rho\colon A \to B$ be a ring homomorphism with image in the centre of B, and let E, F be A-modules. If E is a finitely generated projective A-module, then the canonical homomorphism $B \otimes_A \mathrm{Hom}_A(E, F) \to \mathrm{Hom}_B(B \otimes_A E, B \otimes_A F)$, given by $b \otimes u \mapsto b\,(\mathrm{id}_B \otimes u)$, is bijective.*

If V is a finite-dimensional K-vector space, then V is free with a finite basis, so it is a finitely generated projective K-module by Theorem 3.9. So Theorem 3.10, with $A = K$ and $B = L$, gives a bijection $L \otimes_K \mathrm{Hom}_K(V, W) \to \mathrm{Hom}_L(V_L, W_L)$ for every K-vector space W.

For a K-algebra A with involution σ of $A \otimes A$, let $m_\sigma\colon A \otimes A \to A$ be the product $x \otimes y \mapsto x \cdot_\sigma y$. Define the K-linear map

$$\Phi\colon \mathrm{End}_K(A) \to \mathrm{Hom}_K(A \otimes A, A), \qquad \Phi(D) = D \circ m_\sigma - m_\sigma \circ (D \otimes \mathrm{id} + \mathrm{id} \otimes D).$$

So $\Phi(D)$ is the difference of the two sides of identity (1), and $\mathrm{BSDer}_\sigma(A) = \ker \Phi$. Write Φ_L for the same map for A_L and σ_L, so $\mathrm{BSDer}_{\sigma_L}(A_L) = \ker \Phi_L$.

Proposition 3.11. *Let K be a field, let A be a finite-dimensional K-algebra, and let σ be an involution of $A \otimes A$. Let L be a field extension of K. For $D \in \mathrm{End}_K(A)$ and $c \in L$, let $c \otimes D$ act on A_L by $(c \otimes D)(c' \otimes x) = cc' \otimes D(x)$ for $c' \in L$ and $x \in A$. This rule defines isomorphisms of L-vector spaces*

$$\beta\colon L \otimes_K \mathrm{End}_K(A) \to \mathrm{End}_L(A_L), \qquad \gamma\colon L \otimes_K \mathrm{Hom}_K(A \otimes A, A) \to \mathrm{Hom}_L(A_L \otimes_L A_L, A_L)$$

with $\Phi_L \circ \beta = \gamma \circ (\mathrm{id}_L \otimes \Phi)$. So β and γ carry the exact sequence

$$0 \to L \otimes_K \mathrm{BSDer}_\sigma(A) \to L \otimes_K \mathrm{End}_K(A) \xrightarrow{\mathrm{id}_L \otimes \Phi} L \otimes_K \mathrm{Hom}_K(A \otimes A, A)$$

isomorphically onto the exact sequence

$$0 \to \mathrm{BSDer}_{\sigma_L}(A_L) \to \mathrm{End}_L(A_L) \xrightarrow{\Phi_L} \mathrm{Hom}_L(A_L \otimes_L A_L, A_L).$$

In particular $L \otimes_K \mathrm{BSDer}_\sigma(A) \cong \mathrm{BSDer}_{\sigma_L}(A_L)$ and $\dim_L \mathrm{BSDer}_{\sigma_L}(A_L) = \dim_K \mathrm{BSDer}_\sigma(A)$.

Proof. First, the spaces A and $A \otimes_K A$ are finite-dimensional. So β is an isomorphism by the bijection above with $V = W = A$. Also γ is an isomorphism by the bijection above with $V = A \otimes_K A$ and $W = A$, since $(A \otimes_K A)_L = A_L \otimes_L A_L$.

Second, we check $\Phi_L \circ \beta = \gamma \circ (\mathrm{id}_L \otimes \Phi)$. Both sides are L-linear. So it is enough to evaluate them on $c \otimes D$ and on $(c' \otimes x) \otimes_L (c'' \otimes y)$, for $c, c', c'' \in L$, $D \in \mathrm{End}_K(A)$ and $x, y \in A$. By the construction of σ_L, we have $m_{\sigma_L}\big((c' \otimes x) \otimes_L (c'' \otimes y)\big) = c'c'' \otimes m_\sigma(x \otimes y)$. Hence both sides give

$$cc'c'' \otimes \Big(D\big(m_\sigma(x \otimes y)\big) - m_\sigma\big(D(x) \otimes y + x \otimes D(y)\big)\Big) = cc'c'' \otimes \Phi(D)(x \otimes y).$$

Third, $\ker \Phi = \mathrm{BSDer}_\sigma(A)$. By Theorem 3.8, $\ker(\mathrm{id}_L \otimes \Phi) = L \otimes_K \mathrm{BSDer}_\sigma(A)$. By the second step, β and γ turn $\mathrm{id}_L \otimes \Phi$ into Φ_L. So β maps $L \otimes_K \mathrm{BSDer}_\sigma(A)$ onto $\ker \Phi_L = \mathrm{BSDer}_{\sigma_L}(A_L)$. □

Remark 3.12. The finite-dimensional hypothesis in Proposition 3.11 cannot be dropped. Take $\sigma = -\mathrm{id}$. Then $\sigma^2 = \mathrm{id}$, so σ is an involution of $A \otimes A$, and $x \cdot_\sigma y = 0$ for all x, y. So every K-linear map is a braided-symmetric derivation, both for A and for A_L.

Let $A = K[x]$ and $L = K(t)$, where t is transcendental over K. Define $D' \in \mathrm{End}_L(A_L)$ by $D'(x^i) = t^i x^i$ for all $i \geq 0$. Suppose $D' = \sum_{r=1}^{s} c_r \otimes D_r$ for some $s \geq 1$, with $c_r \in L$ and $D_r \in \mathrm{End}_K(A)$. Since D_r is K-linear and $A = K[x]$, the coefficient of x^i in $D_r(x^i)$ lies in K. So t^i lies in the K-span of $c_1, \ldots, c_s$, for every i. Since t is transcendental over K, the powers $1, t, t^2, \ldots$ are linearly independent over K, so no finite span contains them all. Hence D' is not in the image, and the map of Proposition 3.11 is not onto.

The fixed part $\mathrm{BSDer}_\sigma(A)^\alpha$ is also a kernel, for an automorphism α of A. It is the kernel of the K-linear map

$$\Phi_\alpha \colon \mathrm{End}_K(A) \to \mathrm{Hom}_K(A \otimes A, A) \oplus \mathrm{End}_K(A), \qquad D \mapsto \big(\Phi(D),\, D\alpha - \alpha D\big).$$

Over L, the same map is built from $\alpha_L = \mathrm{id}_L \otimes \alpha$. The isomorphisms β and γ identify it with $\mathrm{id}_L \otimes \Phi_\alpha$. By Theorem 3.8, kernels commute with $L \otimes_K -$. Hence

$$\dim_L \mathrm{BSDer}_{\sigma_L}(A_L)^{\alpha_L} = \dim_K \mathrm{BSDer}_\sigma(A)^\alpha.$$

Here is an example. Let $K = \mathbb{R}$, $A = \mathbb{C}$, and let α be complex conjugation. Then A is commutative and reduced, so Theorem 5.1 below gives

$$\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha = \mathrm{Der}(A)^\alpha.$$

Every $\mathbb{R}$-linear derivation D of $\mathbb{C}$ is zero, since $2iD(i) = D(i^2) = D(-1) = 0$. Now extend to $L = \mathbb{C}$. The algebra $A_L = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ is isomorphic to $\mathbb{C} \times \mathbb{C}$, and α_L becomes the swap of the two factors. Both spaces are zero, as Proposition 3.11 predicts.

4. Twisted involutions and the Leibniz defect

Remark 3.4 treated the identity and the swap, which use no structure of A. This section twists the swap by an involution α of A and treats it with one tool, the Leibniz

defect.

We recall the notions used here. An *automorphism* of A is a bijective K-linear map $\alpha\colon A \to A$ with $\alpha(xy) = \alpha(x)\alpha(y)$ and $\alpha(1) = 1$. An *anti-automorphism* of A is a bijective K-linear map $\alpha\colon A \to A$ with $\alpha(xy) = \alpha(y)\alpha(x)$ and $\alpha(1) = 1$. By $(xy)^T = y^T x^T$, the transpose is an anti-automorphism of $M_n(K)$. In both cases α is *involutive* if $\alpha^2 = \mathrm{id}$. An involutive anti-automorphism is an involution of the ring A in the sense of Herstein [11, §1.2], that is, a map $*$ with $a^{**} = a$, $(a+b)^* = a^* + b^*$ and $(ab)^* = b^* a^*$.

We need $\alpha^2 = \mathrm{id}$ for the following reason. Applying σ_α below twice gives

$$x \otimes y \mapsto \alpha^2(x) \otimes \alpha^2(y),$$

so σ_α is an involution of $A \otimes A$ exactly when α is involutive. When 2 is invertible, an involutive α splits $A = A_+ \oplus A_-$, where $A_\pm = \{x \in A : \alpha(x) = \pm x\}$. Indeed, for $x \in A$,

$$x = \tfrac{1}{2}\big(x + \alpha(x)\big) + \tfrac{1}{2}\big(x - \alpha(x)\big).$$

We call the elements of A_+ *symmetric* and the elements of A_- *skew*.

Let α be involutive, of either kind. Then $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$ is a K-linear involution of $A \otimes A$. So σ_α defines a symmetrized product as in Section 2. The product it defines is

$$x \cdot_{\sigma_\alpha} y = xy + \alpha(y)\alpha(x). \tag{2}$$

For $\alpha = \mathrm{id}$ the product μ_{σ_α} is the Jordan product. If α is an automorphism, then $\alpha(y)\alpha(x) = \alpha(yx)$. If α is an anti-automorphism, then $\alpha(y)\alpha(x) = \alpha(xy)$.

Definition 4.1. The *Leibniz defect* of a linear map $D\colon A \to A$ is the bilinear map $E_D\colon A \times A \to A$ given, for $x, y \in A$, by

$$E_D(x, y) = D(xy) - D(x)y - xD(y).$$

The name is ours, and the map is classical: Cusack [9, §2] writes it $d(a,b) = D(ab) - aDb - (Da)b$ and studies Jordan derivations through it. We recall the Hochschild complex of A with coefficients in A [24, §9.1]. An *n-cochain* is a K-multilinear map from A^n to A. So a 0-cochain is an element of A, and a 1-cochain is a linear map $D\colon A \to A$. The *coboundary* δ sends each n-cochain to an $(n+1)$-cochain, and $\delta \circ \delta = 0$. An *n-cocycle* is an n-cochain f with $\delta f = 0$. The n-th *Hochschild cohomology* $HH^n(A, A)$ is the space of n-cocycles modulo the images δg of the $(n-1)$-cochains g [24, §9.1].

On a 1-cochain D the coboundary is $(\delta D)(x, y) = xD(y) - D(xy) + D(x)y$ [12, §2, Definition 2.1] [24, §9.1]. So $E_D = -\delta D$. Thus D is a derivation exactly when $E_D = 0$, so derivations are the 1-cocycles. Since $\delta \circ \delta = 0$, the defect is a 2-cocycle: for all $a, b, c \in A$,

$$aE_D(b,c) - E_D(ab,c) + E_D(a,bc) - E_D(a,b)c = 0. \tag{3}$$

This says that E_D is a cocycle of the Hochschild complex [24, §9.1].

The defect also describes the other classes of Section 2. Expanding $D(x \circ y)$ shows that D is a Jordan derivation exactly when $E_D(x,y) + E_D(y,x) = 0$ for all x, y. In the

same way, D is a Lie derivation exactly when $E_D(x,y) = E_D(y,x)$ for all x, y.

Proposition 4.2. *Let A be a unital associative K-algebra, let α be an involutive automorphism or anti-automorphism of A, and let $D\colon A \to A$ be a linear map with $D\alpha = \alpha D$.*

(i) If α is an automorphism, then $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $E_D(x,y) + \alpha(E_D(y,x)) = 0$ for all $x, y \in A$.

(ii) If α is an anti-automorphism, then $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $E_D(x,y) + \alpha(E_D(x,y)) = 0$ for all $x, y \in A$.

In both cases every derivation that commutes with α lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$.

Proof. Write $E = E_D$. By Remark 3.4, $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ means $D(x \cdot_{\sigma_\alpha} y) = D(x) \cdot_{\sigma_\alpha} y + x \cdot_{\sigma_\alpha} D(y)$ for all x, y. We compute the difference of the two sides in each case.

For (i), the product (2) is $x \cdot_{\sigma_\alpha} y = xy + \alpha(yx)$. Since $D\alpha = \alpha D$, the left side is $D(xy) + \alpha(D(yx))$. The right side is $D(x)y + \alpha(yD(x)) + xD(y) + \alpha(D(y)x)$. We group the terms without α and the terms with α. The first group gives $D(xy) - D(x)y - xD(y) = E(x,y)$. Since α is linear, the second group gives $\alpha\big(D(yx) - D(y)x - yD(x)\big) = \alpha(E(y,x))$. So the left side minus the right side is $E(x,y) + \alpha(E(y,x))$, which proves (i).

The proof of (ii) follows the same steps as the proof of (i). Here the product (2) is $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$. Since $D\alpha = \alpha D$, the left side is

$$D(xy) + \alpha(D(xy)),$$

and the right side is

$$D(x)y + \alpha(D(x)y) + xD(y) + \alpha(xD(y)).$$

The same grouping gives the difference

$$E(x,y) + \alpha(E(x,y)),$$

which proves (ii).

If D is a derivation, then $E = 0$, and both conditions hold. □

We now read the two conditions, assuming that 2 is invertible.

In case (ii), the equation $E_D + \alpha(E_D) = 0$ says $\alpha(E_D) = -E_D$. So E_D takes values in A_-, the skew elements of α. In case (i), we write $E_D = E_\mathrm{s} + E_\mathrm{a}$. Here $E_\mathrm{s}(x,y) = \frac{1}{2}(E_D(x,y) + E_D(y,x))$ is symmetric. And $E_\mathrm{a}(x,y) = \frac{1}{2}(E_D(x,y) - E_D(y,x))$ is antisymmetric.

We swap x and y in the condition of (i). Then we take the sum and the difference of the two equations. So the condition holds exactly when E_s takes values in A_- and E_a takes values in A_+. Now let $\alpha = \mathrm{id}$. Then $A_- = 0$ and $A_+ = A$, so the condition says $E_\mathrm{s} = 0$. This is the Jordan condition, so Proposition 4.2(i) recovers the swap case of Remark 3.4.

Remark 4.3. Let α be an involutive automorphism of A, and let $D \in \mathrm{Der}(A)$. Then $E_D = 0$. By (2), the two sides of the σ_α-identity differ by

$$D(\alpha(yx)) - \alpha(D(yx)).$$

Since A is unital, the products yx span A. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $D\alpha = \alpha D$, and then $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$.

Next we look at the inner derivations ad_g in this class. The condition $\mathrm{ad}_g\,\alpha = \alpha\,\mathrm{ad}_g$ means

$$[g - \alpha(g), \alpha(x)] = 0 \qquad \text{for all } x.$$

Since α is bijective, this holds if and only if $g - \alpha(g) \in Z(A)$. Let $\mathrm{Inn}_\alpha(A)$ be the space of the maps ad_g with $g - \alpha(g) \in Z(A)$. These are the inner derivations in $\mathrm{BSDer}_{\sigma_\alpha}(A)$, so $\mathrm{Inn}_\alpha(A) = \mathrm{Inn}(A) \cap \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. The quotient $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha / \mathrm{Inn}_\alpha(A)$ is the space of *twisted outer derivations*. For $\alpha = \mathrm{id}$ the quotient is $\mathrm{Der}(A)/\mathrm{Inn}(A) = HH^1(A, A)$, by Lemma 6.5.

Cusack [9, Corollary 5] and Brešar [5, Theorem 1] proved that a Jordan derivation of a 2-torsion free semiprime ring is a derivation.

Theorem 4.4 ([5, Theorem 1])**.** *Let R be a 2-torsion free semiprime ring. Then every Jordan derivation $D\colon R \to R$ is a derivation.*

Corollary 4.5. *Let A be a 2-torsion-free semiprime K-algebra with an involutive automorphism α. Then $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \cap \mathrm{JDer}(A) = \mathrm{Der}(A)^\alpha$, where $\mathrm{Der}(A)^\alpha$ is the set of derivations commuting with α.*

Proof. Every derivation is a Jordan derivation by Remark 3.4. With Proposition 4.2, this gives $\mathrm{Der}(A)^\alpha \subseteq \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \cap \mathrm{JDer}(A)$. Conversely, every Jordan derivation of A is a derivation by Theorem 4.4, which applies since a K-linear map is additive. So a map in the intersection is a derivation that commutes with α. □

5. Reduced commutative algebras and twisted derivations

For commutative algebras the Jordan hypothesis in Corollary 4.5 can be dropped. Recall from Section 2 that a reduced commutative algebra has no nonzero nilpotent element.

Theorem 5.1. *Let A be a commutative reduced K-algebra with 2 invertible in K, and let α be an involutive automorphism of A. Then $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha = \mathrm{Der}(A)^\alpha$.*

Proof. Every derivation commuting with α lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 4.2. Conversely, let $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. Write $E = E_D$. Since A is commutative, E is symmetric. So E takes values in A_- by Proposition 4.2(i).

Since D commutes with α, it maps each of A_+ and A_- into itself. So if x and y lie in the same part, then $E(x, y)$ lies in

$$A_+ \cap A_- = 0.$$

Now let $x \in A_+$ and $y \in A_-$, and put $z = E(x, y)$. Then z lies in A_-, and so do xy, zx and xz, while zy lies in A_+. Apply the cocycle identity (3) to $(a, b, c) = (z, x, y)$:

$$zE(x, y) - E(zx, y) + E(z, xy) - E(z, x)y = 0.$$

The pairs (zx, y) and (z, xy) lie in A_-, so E vanishes on them, and we get $z^2 =$

$E(z,x)y$. Apply (3) to $(a,b,c) = (x,z,y)$:

$$xE(z,y) - E(xz,y) + E(x,zy) - E(x,z)y = 0.$$

The pairs (z,y) and (xz,y) lie in A_-, and the pair (x,zy) lies in A_+, so E vanishes on all three and $E(x,z)y = 0$. Since E is symmetric, $z^2 = E(z,x)y = E(x,z)y = 0$. So $z = 0$ because A is reduced. Since E is bilinear and symmetric and $A = A_+ \oplus A_-$, we get $E = 0$. Hence D is a derivation. □

The mixed-parity step uses the cocycle identity twice. Example 5.2 shows that the reduced hypothesis cannot be dropped, and Example 5.3 shows the same for the invertibility of 2.

Example 5.2. Let $A = \mathbb{F}_5[u,v]/(u^2,v^2)$. It has basis $\{1,u,v,uv\}$. Let α fix u and send v to $-v$. Let D send uv to uv and the other three basis elements to 0. Then D commutes with α. It satisfies the Leibniz rule for $x \cdot_{\sigma_\alpha} y$ on all basis pairs. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. But D is not a derivation, since

$$D(uv) = uv \qquad \text{while} \qquad D(u)v + uD(v) = 0.$$

Direct computation on the four basis elements verifies all three statements. Here $z = E(u,v) = uv$ is a nonzero element with $z^2 = 0$.

Example 5.3. Let $A = \mathbb{F}_2 \times \mathbb{F}_2$, which is commutative and reduced, and let α swap the two factors. Put $p = (1,0)$, $q = (0,1)$ and $e = (1,1)$, and let D be the linear map with $D(p) = D(q) = e$, so that $D(e) = 0$. Then D commutes with α. On the basis pairs (p,p), (q,q) and (p,q), both sides of the Leibniz rule for $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$ are zero. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. But D is not a derivation: $D(p) = D(p^2) = 2pD(p) = 0$ would hold for a derivation, while $D(p) = e$.

Our second natural candidate for the twisted class is the space $\mathrm{Der}_\alpha(A)$ of α-derivations. It was defined for automorphisms in Section 2. It differs from $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ in both directions, and one algebra shows both failures.

Proposition 5.4. *Let $A = \mathbb{F}_3[\varepsilon]/(\varepsilon^3)$, and let α be the automorphism with $\alpha(\varepsilon) = -\varepsilon$. Then*

$$\mathrm{Der}_\alpha(A) \not\subseteq \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \qquad \textit{and} \qquad \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \not\subseteq \mathrm{Der}_\alpha(A).$$

Proof. The algebra A has basis $\{1,\varepsilon,\varepsilon^2\}$. The map α fixes 1 and ε^2 and sends ε to $-\varepsilon$. So it is an involutive automorphism with $A_+ = \mathbb{F}_3 1 + \mathbb{F}_3\varepsilon^2$ and $A_- = \mathbb{F}_3\varepsilon$. Since A is commutative, E_D is symmetric for every linear map D. Let D be a linear map with $D\alpha = \alpha D$. By Proposition 4.2(i), D lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)$ exactly when E_D takes values in A_-. We give one map for each non-inclusion. Both maps are diagonal on the basis, so both commute with α.

For the first non-inclusion, let $D(a + b\varepsilon + c\varepsilon^2) = b\varepsilon + c\varepsilon^2$. We check that D is an α-derivation on pairs of basis elements. Pairs that contain 1 give 0 on both sides, since $D(1) = 0$ and $\alpha(1) = 1$. For the pair $(\varepsilon,\varepsilon)$, the left side is $D(\varepsilon^2) = \varepsilon^2$. The right side is $D(\varepsilon)\alpha(\varepsilon) + \alpha(\varepsilon)D(\varepsilon) = -2\varepsilon^2 = \varepsilon^2$. The remaining pairs have product 0, and each term on the right side is a multiple of $\varepsilon^3 = 0$. So $D \in \mathrm{Der}_\alpha(A)$. On the other hand, $E_D(\varepsilon,\varepsilon) = D(\varepsilon^2) - 2\varepsilon D(\varepsilon) = \varepsilon^2 - 2\varepsilon^2 = 2\varepsilon^2$. This element is nonzero and lies in A_+, so it does not lie in A_-. Hence $D \notin \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$.

The second non-inclusion uses a derivation.

Let d be the linear map

$$d(a + b\varepsilon + c\varepsilon^2) = b\varepsilon + 2c\varepsilon^2.$$

It is a derivation, since $d(\varepsilon^i) = i\varepsilon^i$ and exponents add under multiplication. So $E_d = 0$, and $d \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. On the other hand,

$$d(\varepsilon^2) - d(\varepsilon)\alpha(\varepsilon) - \alpha(\varepsilon)d(\varepsilon) = 2\varepsilon^2 + 2\varepsilon^2 = \varepsilon^2 \neq 0.$$

Hence $d \notin \mathrm{Der}_\alpha(A)$. □

The two classes can also meet, and the next example shows both cases.

Example 5.5. Let $A = M_2(\mathbb{F}_3)$ and let α be the transpose, an anti-automorphism. Then $\mathrm{Der}_\alpha(A)$ is not defined, since it needs an automorphism. Take an inner derivation ad_g with g skew for the transpose. It commutes with α and has $E = 0$, so it lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 4.2.

Now let $A = K[\varepsilon]/(\varepsilon^2)$ with $\alpha(\varepsilon) = -\varepsilon$. The map $D(a+b\varepsilon) = b\varepsilon$ is a derivation and an α-derivation, and it commutes with α. So it lies in both $\mathrm{Der}_\alpha(A)$ and $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 4.2.

6. The antisymmetrized product and Lie derivations

This section replaces the sum $\mu + \mu \circ \sigma$ by the difference $\mu - \mu \circ \sigma$. For the swap the difference is the commutator, and its derivations are the Lie derivations.

Definition 6.1. The *σ-antisymmetrized product* on a K-algebra A is $x \cdot_\sigma^- y = \mu(x \otimes y) - \mu(\sigma(x \otimes y))$, where σ is an involution of $A \otimes A$. A *braided-antisymmetric derivation* is a K-linear map $D\colon A \to A$ with $D(x \cdot_\sigma^- y) = D(x) \cdot_\sigma^- y + x \cdot_\sigma^- D(y)$.

We now identify these derivations for the swap. It also gives the easy half of the theorems of Martindale and Brešar.

Proposition 6.2. *Let A be an associative K-algebra.*

(i) The braided-antisymmetric derivations for τ in the sense of Definition 6.1 are exactly the Lie derivations of A.
(ii) $\mathrm{Der}(A) \subseteq \mathrm{LDer}(A)$.
(iii) Let $d \in \mathrm{Der}(A)$, and let $t\colon A \to Z(A)$ be linear with $t([x,y]) = 0$ for all x, y. Then $d + t \in \mathrm{LDer}(A)$.
(iv) In (iii), if $t(1) \neq 0$, then $d + t \notin \mathrm{Der}(A)$.

Proof. We check the four parts in order. (i) For the swap, $x \cdot_\tau^- y = [x, y]$, so the two identities are the same. (ii) For $D \in \mathrm{Der}(A)$, $D([x,y]) = D(x)y + xD(y) - D(y)x - yD(x) = [D(x), y] + [x, D(y)]$. (iii) By (ii), $d([x,y]) = [d(x), y] + [x, d(y)]$, and $t([x,y]) = 0$. Since $t(x)$ and $t(y)$ are central, $[t(x), y] = 0 = [x, t(y)]$. Adding gives the Lie derivation identity for $d + t$. (iv) Every derivation vanishes at 1, as recalled in Section 2, while $(d+t)(1) = t(1) \neq 0$. □

We now twist the swap as in Section 4. The twisted antisymmetrized product is $x \cdot_{\sigma_\alpha}^- y = xy - \alpha(y)\alpha(x)$.

Proposition 6.3. *Let A be a unital associative K-algebra, let α be an involutive automorphism or anti-automorphism of A, and let $D\colon A \to A$ be a linear map with $D\alpha = \alpha D$.*

(i) If α is an automorphism, then D is a braided-antisymmetric derivation for σ_α if and only if $E_D(x,y) - \alpha(E_D(y,x)) = 0$ for all $x, y \in A$.
(ii) If α is an anti-automorphism, then D is a braided-antisymmetric derivation for σ_α if and only if $E_D(x,y) - \alpha(E_D(x,y)) = 0$ for all $x, y \in A$.

Proof. Write $E = E_D$. For (i), the product is $x \cdot_{\sigma_\alpha}^{-} y = xy - \alpha(yx)$. Since $D\alpha = \alpha D$, the left side of the identity is $D(xy) - \alpha(D(yx))$. The right side is $D(x)y - \alpha(yD(x)) + xD(y) - \alpha(D(y)x)$. The terms without α give $E(x,y)$ in the difference. Since α is linear, the terms with α give $-\alpha\big(D(yx) - yD(x) - D(y)x\big) = -\alpha(E(y,x))$. So the left side minus the right side is $E(x,y) - \alpha(E(y,x))$. For (ii), the product is $x \cdot_{\sigma_\alpha}^{-} y = xy - \alpha(xy)$. The same steps give the difference $E(x,y) - \alpha(E(x,y))$. □

Take $\alpha = \mathrm{id}$ in (i). Then the condition says that E_D is symmetric. This is the condition for a Lie derivation from Section 4.

Now let 2 be invertible. In case (ii), the condition says that E_D takes values in A_+. In case (i), write $E_D = E_\mathrm{s} + E_\mathrm{a}$ as in Section 4. The condition says that E_s takes values in A_+ and E_a takes values in A_-. To see this, swap x and y in (i), then add the two equations and subtract them. The parities are the opposite of those for the symmetrized product in Proposition 4.2.

So inner derivations lie in both spaces. Each ad_g is a Lie derivation by Proposition 6.2(ii), and a Jordan derivation by Remark 3.4. Full matrix algebras show that part (iii) gives maps outside $\mathrm{Der}(A)$. We use one result on Jordan derivations of matrix rings. It is stated for a unital associative ring and a 2-torsion free bimodule [1, §2].

Theorem 6.4 ([1, Theorem 3.1])**.** *Let A be a unital associative ring, let M be a 2-torsion free A-bimodule, and let $n \geq 2$. Then every Jordan derivation $D\colon M_n(A) \to M_n(M)$ is a derivation.*

We also use a description of the first Hochschild cohomology. It compares the twisted outer derivations of Remark 4.3 with the ordinary outer derivations, and it gives HH^1 of the matrix algebra in Example 6.9. Let R be a k-algebra and M an R-bimodule. Weibel [24, §9.2] writes $\mathrm{Der}_k(R,M)$ for the k-derivations. Weibel writes $\mathrm{PDer}_k(R,M)$ for the principal ones $f_m(r) = rm - mr$.

Lemma 6.5 ([24, Lemma 9.2.1])**.** *Let R be a k-algebra and M an R-bimodule. Then $HH^1(R,M) \cong \mathrm{Der}_k(R,M)/\mathrm{PDer}_k(R,M)$.*

A nonzero ring R is *simple* if (0) and R are the only ideals in R, and a commutative ring is simple if and only if it is a field [17, §1].

The next theorem shows that $M_n(K)$ is simple. Example 6.9 needs this twice: to apply the theorem of Martindale, and to see that $M_n(K)$ is separable.

Theorem 6.6 ([17, Theorem 3.1])**.** *Let R be a ring and $M_n(R)$ be the ring of $n \times n$ matrices over R. Then any ideal I of $M_n(R)$ has the form $M_n(\mathfrak{m})$ for a uniquely determined ideal $\mathfrak{m}$ of R. In particular, if R is a simple ring, so is $M_n(R)$.*

Theorem 6.7 ([18, Theorem 2])**.** *Let D be a Lie derivation of a primitive ring R into itself, where R contains a nontrivial idempotent and the characteristic of R is not 2. Then D is of the form $d + t$, where d is an ordinary derivation of R into a primitive*

ring $\hat{R}$ containing R and t is an additive mapping of R into the center of $\hat{R}$ that maps commutators into zero.

Martindale remarks after Theorem 6.7 that, if R is simple, then d maps R into itself and t maps R into the center of R [18, p. 187].

Theorem 6.8 ([12, Theorem 4.1])**.** *A necessary and sufficient condition for an algebra $\mathfrak{A}$ to be separable is that $H^1(\mathfrak{A}, \mathfrak{M}) = \{0\}$ for every two-sided $\mathfrak{A}$-module $\mathfrak{M}$.*

We now apply Theorems 6.4, 6.6, 6.7 and 6.8 to $M_n(K)$.

Example 6.9 (Full matrix algebras)**.** Let $A = M_n(K)$ with $n \geq 2$, and let e_{ij} be the matrix unit with entry 1 in position (i, j) and 0 elsewhere. If $\operatorname{char} K \neq 2$, every Jordan derivation of $M_n(K)$ is a derivation by Theorem 6.4 applied to the ring K and the K-bimodule K, since K is then 2-torsion free. So $\mathrm{BSDer}_\tau(A) = \mathrm{JDer}(A) = \mathrm{Der}(A)$ in that case.

Let $\theta(x) = \operatorname{tr}(x)I$. Its values are central, and $\theta([x, y]) = (\operatorname{tr}(xy) - \operatorname{tr}(yx))I = 0$. By Proposition 6.2(iii), $\mathrm{ad}_g + c\,\theta$ is a Lie derivation for every $g \in A$ and $c \in K$. Since $\theta(e_{11}) = I$ while $\theta(e_{11})e_{11} + e_{11}\theta(e_{11}) = 2e_{11}$ and $\theta(e_{11}^2) = \theta(e_{11})$, the map θ is not a derivation in any characteristic.

Now let $n \geq 2$ and $\operatorname{char} K \neq 2$. The ring $M_n(K)$ is simple by Theorem 6.6, since a field is simple. It is primitive, because K^n is a faithful simple left module over it, and e_{11} is a nontrivial idempotent. By Theorem 6.7 and the remark after it, every Lie derivation D of $M_n(K)$ has the form $D = d + t$, where d is an additive derivation of $M_n(K)$ into itself and t is an additive map into $K \cdot I$ that vanishes on commutators.

The maps d and t are K-linear. Indeed, for $\lambda \in K$ we have $d(\lambda x) = d(\lambda I)x + \lambda d(x)$, and $d(\lambda I)$ is central because $[d(\lambda I), x] = d([\lambda I, x]) - [\lambda I, d(x)] = 0$. So $d(\lambda I) = \nu(\lambda)I$ for some $\nu(\lambda) \in K$. Since D is K-linear, $\nu(\lambda)x = \lambda t(x) - t(\lambda x)$ lies in $K \cdot I$ for every x. Taking $x = e_{11}$ gives $\nu(\lambda) = 0$.

Next, $M_n(K)$ is separable, since $M_n(K) \otimes_K F = M_n(F)$ is simple for every extension field F by Theorem 6.6. By Theorem 6.8, $H^1(M_n(K), M_n(K)) = 0$, so every K-linear derivation is inner, and d is inner. Hence $\mathrm{Der}(A) = \mathrm{Inn}(A)$, and every Lie derivation of A has the form $\mathrm{ad}_g + t$ with $t\colon A \to K \cdot I$ linear and $t([A, A]) = 0$. Such a map t is the same as a K-linear map $A/[A, A] \to K$, and $A/[A, A] = HH_0(A, A)$ [24, §9.1]. A map t of this kind that is also a derivation is zero. Indeed $t(e_{ii}) = t(e_{ii}^2) = 2t(e_{ii})e_{ii}$ forces $t(e_{ii}) = 0$ when $n \geq 2$, and $t(e_{ij}) = 0$ for $i \neq j$ because $e_{ij} = [e_{ii}, e_{ij}]$. So t is determined by $\mathrm{ad}_g + t$: if $\mathrm{ad}_g + t = \mathrm{ad}_h + t'$, then $t - t' = \mathrm{ad}_{h-g}$ is such a derivation, hence zero. Let $\omega\colon \mathrm{LDer}(A) \to \mathrm{Hom}_K(HH_0(A, A), K)$ send $\mathrm{ad}_g + t$ to t. The map ω is surjective, since each t is a Lie derivation by Proposition 6.2(iii), and its kernel is $\mathrm{Inn}(A) = \mathrm{Der}(A)$. So the sequence

$$0 \to \mathrm{Der}(A) \to \mathrm{LDer}(A) \xrightarrow{\omega} \mathrm{Hom}_K\big(HH_0(A, A), K\big) \to 0$$

is exact. The map θ above is one element of the right-hand space.

The swap gives $\mathrm{JDer}(A)$ through $\mu + \mu\circ\tau$ and $\mathrm{LDer}(A)$ through $\mu - \mu\circ\tau$. So μ is half the sum of these two products when 2 is invertible. So a map that is both a Jordan and a Lie derivation is a derivation. On $M_n(K)$ this gives $\mathrm{JDer}(A) \cap \mathrm{LDer}(A) = \mathrm{Der}(A)$ even though $\mathrm{LDer}(A)$ is larger.

For triangular algebras the Lie side has a central term.

A Lie derivation is *proper* when it is the sum of a derivation and a linear map into

the centre [8, §1]. The extra map vanishes on commutators, so a proper Lie derivation has the standard form of Section 1.

The next theorem uses the algebra $\mathfrak{A} = \mathrm{Tri}(A, M, B)$, and the projections π_A and π_B onto the diagonal blocks.

The algebra $W(X)$ is the smallest subalgebra of X that satisfies seven closure conditions of Cheung [8, Proposition 10]. We list them for a subalgebra V of X. Here F is a polynomial in one variable over the base ring, and F' is its derivative.

(i) V contains all commutators $[x, y]$ with $x, y \in X$.
(ii) If $b \in X$ and $F'(b) = 0$, then $F(b) \in V$.
(iii) If $b \in X$, $F(b) \in V$ and $F'(b)$ is invertible, then $b \in V$.
(iv) V contains all idempotents of X.
(v) V contains all elements x^p, where $x \in X$ and $p \geq 0$ is the characteristic of X.
(vi) If $a \in X$ has an inverse a^{-1} in X and $a^{-1} \in V$, then $a \in V$.
(vii) If $b \in X$ is invertible and $b^k \in V$ for some positive integer k, then $kb \in V$.

Theorem 6.10 ([8, Theorem 11])**.** *Let $\mathfrak{A} = \mathrm{Tri}(A, M, B)$. If the following two conditions hold, then every Lie derivation of $\mathfrak{A}$ is proper.*

(I) M is a faithful left A-module and $Z(B) = \pi_B(Z(\mathfrak{A}))$; or M is a faithful left A-module and $A = W(A)$; or every Lie derivation of A is proper and $A = W(A)$.
(II) M is a faithful right B-module and $Z(A) = \pi_A(Z(\mathfrak{A}))$; or M is a faithful right B-module and $B = W(B)$; or every Lie derivation of B is proper and $B = W(B)$.

Li and Wei [16, §3] extend this analysis to generalized matrix algebras. A generalized matrix algebra [16, §2.1] is built from two algebras A and B and two bimodules ${}_AM_B$ and ${}_BN_A$. It also has two bimodule maps

$$M \otimes_B N \to A, \qquad N \otimes_A M \to B.$$

These data form a Morita context [16, §2.1]. The bimodule N may be zero, and then the algebra is triangular [16, §2.1 and Corollary 3.6].

Corollary 6.11. *Let $T = \mathrm{Tri}(A, M, B)$ satisfy conditions (I) and (II) of Theorem 6.10. Then every braided-antisymmetric derivation of T for τ has the standard form $d + t$, and it is a derivation only if $t(1) = 0$.*

Proof. By Proposition 6.2(i), the maps in question are the Lie derivations of T, which are proper by Theorem 6.10. The last claim is Proposition 6.2(iv). □

Corollary 3.7 says that on such a triangular algebra the symmetrized product gives only derivations. Example 6.12 shows that the antisymmetrized product can add a central term. Take $T_2(K) = \mathrm{Tri}(K, K, K)$ over a field with $\mathrm{char}\, K \neq 2$. Here $M = K$ is faithful on both sides. Also $Z(T_2(K)) = KI$ projects onto K. So Corollaries 3.7 and 6.11 apply.

Example 6.12 (Upper triangular 2×2 matrices)**.** Let $T = T_2(\mathbb{F}_3)$. Every additive map $T \to T$ is $\mathbb{F}_3$-linear. So it is one of the 3^9 maps given by a 3×3 matrix over $\mathbb{F}_3$.

An exhaustive computer check over all of them confirms Corollaries 3.7 and 6.11. Every Jordan derivation is a derivation. Every Lie derivation equals $\mathrm{ad}_x + t$ for some $x \in T$ and some central-valued t that vanishes on commutators.

The term t is needed. Define θ by

$$\theta\begin{pmatrix} a & m \\ 0 & b \end{pmatrix} = aI.$$

It has central values and vanishes on commutators, whose diagonal is zero. So it is a Lie derivation. It is not a derivation, since

$$\theta(e_1^2) = I \qquad \text{while} \qquad \theta(e_1)e_1 + e_1\theta(e_1) = 2e_1.$$

So on $T_2(\mathbb{F}_3)$, as on $M_n(K)$, we have JDer = Der, while LDer is strictly larger.

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